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ORIGINAL RESEARCH REPORT



Prenatal noise and restraint stress interact to alter exploratory behavior and balance in juvenile rats, and mixed stress reverses these effects

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ABSTRACT

We aimed to investigate in adolescent rats the individual and combined effects of prenatal noise and restraint stress on balance control, exploration, locomotion and anxiety behavior. Three groups of pregnant rats were exposed to daily repeated stress from day 11 to day 19 of pregnancy: 3 min noise (Noise Stress, NS); 10 min restraint (restraint stress, RS); or 3 min noise followed by 10 min restraint (mixed stress, MS). On postnatal days (PND) 44, 45 and 46, four groups of male rats (Control, NS, RS; MS; 16 rats each), were tested as follows: (1) beam walking (BW), (2) open field (OF) and (3) elevated plus maze (EPM). Our results show that the NS group had significantly impaired balance control, locomotion and both horizontal and vertical exploration ($p < .01$ for all measures). The RS group showed only a decrease in vertical exploration ($p < .05$). In contrast, locomotion and balance were not affected in the MS group (OF: crossed squares: $p = .34$, missteps: $p = .18$). However, MS rats exhibited significantly higher anxiety levels (less time in EPM open arms: $p < .05$), and took more time to complete BW: $p < .05$). Hence, combined prenatal stressors exert non-additive effects on locomotion, exploration and balance control, but induce greater anxiety through additive effects. Terminal plasma ACTH concentration was increased by prenatal stress, especially noise, which group had the largest adrenal glands. Overall, contrary to expectation, combined prenatal stressors can interact to increase anxiety level, but diminish alteration of exploration, locomotion and impaired balance control, which were strongly induced by noise stress.

Lay summary: Experience of stress in pregnancy can have negative effects on the offspring that are long-lasting. Here, we used laboratory rats to see whether repeated episodes of exposure to loud noise or preventing free movement, alone or together, during pregnancy had different effects on behaviors of the adolescent offspring. Using standard tests, we found the prenatal stresses caused the offspring to be anxious, and not to balance when moving around as well as normal offspring; the degree of impairment depended on the type of stress – loud noise exposure had the greatest effects, but if the stresses were combined the effects were not worse. The results point to the need to aim to avoid stress in pregnant women.

ARTICLE HISTORY

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Anxiety; balance; beam walking; locomotion; noise; prenatal stress

Introduction

Over the past few decades, many studies have focused on the short-term effects of prenatal stress on fetal development in both animal and human models (Cao-Lei et al., 2015; Zeng et al., 2015). Indeed, it has been reported that prenatal exposure to stress causes neurophysiological (Caspi et al., 2010), cognitive and behavioral disorders (Cabrera et al., 1999) in offspring, such as spatial memory disorders (Benoit et al., 2015) development of anxiety-or depression-like behavior (Basner et al., 2014; Weinstock, 2008).

It has been hypothesized that epigenetic mechanisms represent the interface through which the environment acts to modify the expression of genes and, at the same time, induces behavioral changes (Zheng et al., 2016). Thus, major traumatic events in life such as loss of a spouse or experiencing

armed conflict (Appart et al., 2016), has been associated with post-traumatic stress disorder (PTSD) (Yehuda et al., 2005) and increased incidence of schizophrenia in the offspring of pregnant women, particularly those traumatized in the first trimester of pregnancy (van Os & Selten, 1998). Indeed, several studies on rats have concluded that prenatal stress alters memory, learning and exploration, with development of anxiety- or depression-like behavior (Babri et al., 2014; Cabrera et al., 1999). However, results of studies on prenatal stress in animals are sometimes contradictory, with Meek et al. (2000) showing that 12-day-old mice born to mothers stressed during the last week of gestation exhibited more intense exploratory capacity than the control.

Several important questions arise from these studies, such as how does exposure to events that occur during the

perinatal period affect behavior (exploration, locomotion and anxiety-like behavior) later in life?; When various different stressors are applied simultaneously, is the effect similar to a sum-of-effects of the individually-applied stressors? and can these stressors act synergistically or antagonistically to modify behavior in juvenile rats? In order to answer these questions, we investigated the impact of prenatal exposure to two types of stressor, namely sound stress and restraint stress, alone and as a combination of both. Environmental noise (vehicles, work environment) is usually not threatening, but it may represent a considerable subjective annoyance and act as a stressor, leading to psychosomatic disturbances (Haüsler 2004). For this reason, we aimed to investigate the effects of chronic short exposure to high decibel noise of the mother during pregnancy to mimic natural exposure to a high level environmental noise (e.g. vehicle horns, aircraft noise). Here, restraint stress was selected as a model of emotional prenatal stress on the basis of previous studies reporting that gestational restraint stress induces long-lasting neurobiological and behavioral alterations (Maccari & Morley-Fletcher, 2007).

We evaluated anxiety like-behavior, exploratory behavior, locomotor performance and balance control in adolescent rats. Balance control appears to have received little attention in the field of stress, yet prenatal stress exposure is likely to significantly influence balance control. Nazeri et al. (2015) reported that balance and motor function were impaired in prenatally stressed juvenile female rats. A study on non-human primates found that strong exposure to prenatal stress resulted in reduced motor activity and poorer balance in the exposed offspring (Schneider & Coe, 1993). In particular, the effect of simultaneous exposure to different stressors during intrauterine life on balance control has not been studied.

Thus, a novel aspect of the present study is the testing of locomotor functions in the form of balance control. We also investigated whether prenatal noise and restraint stress interact and how this interaction impacts exploration and locomotion, as well as on balance control of juvenile rats. Overall, the goal of the present study was to determine the extent to which prenatal noise and restraint stress, applied individually, impacted exploratory behavior, locomotor performance and balance control functioning in juvenile male rats, and how these prenatal stressors interact to modify behavior when applied together.

Methods

Animals

Test subjects were 24 90-day-old virgin female rats (Wistar albino strain provided by Pasteur Institute-Algiers, Algeria), weighing 200 ± 20 g at the beginning of the study. Rats were housed in PVC cages in a colony room (Department of Biology, University of Annaba, Algeria) and kept under standard laboratory conditions of $25 \pm 2^\circ\text{C}$, 50% relative humidity and artificial lighting, 12 h light/12 h dark starting at 07h00. Food (standard rat chow from ONAB-El Kseur, Béjaïa, Algeria), and water were provided *ad libitum*. Unnecessary manipulations (i.e. apart from cage cleaning) were avoided.

Experimental protocols

In order to synchronize estrous cycles in the female rats, two males of the same strain and four female rats were put into a partitioned cage with no physical contact for 7 days, allowing exposure to male pheromones. Males and females were then separated for 7 days before pairs of females were caged with one of the males for copulation. Vaginal smears were analyzed daily and observation of sperm was considered day 0 of pregnancy. Males were then removed and pregnant females were separated into experimental groups with two rats per cage: noise stress (NS), $n=6$; restraint stress (RS) $n=6$; mixed stress (MS) $n=6$; and control (C) $n=8$. Pregnant females were separated into individual cages on day 17 as the end of pregnancy approached.

Seventy-two hours after delivery, prenatally stressed and control offspring were sexed and weighed (data not shown), before being returned to their own dams to nurse undisturbed under the standard laboratory conditions described above. After day 30 post-partum, female rats in the progeny were excluded from our experiments (because the aim of this study was not investigation of sex-biased responses to stressors) and 16 males were selected randomly from each group of surviving rats (Control: $n=16$, NS: $n=16$, RS: $n=16$, and MS: $n=16$). All 16 rats of each group were subjected to behavioral tests to assess anxiety levels, locomotor and exploratory behavior and balance ability. It is important to note that all rats were kept undisturbed until post-natal day (PND) 30 before being divided into four groups of 16 rats (C, NS, RS and MS groups; eight rats per cage). Behavioral test sessions were performed with 24-h intervals between each test to avoid acute stress-induced interference with the outcome of the subsequent test. Blood sampling was performed to measure plasma adrenocorticotrophic hormone (ACTH) concentrations after 48 h of behavioral assessment to assess whether hormonal responses had returned to baseline levels. All experimental procedures described in this work are in accordance with the Guide of Care and Use of Laboratory Animals of the University of Badji Mokhtar Annaba, Algeria.

Maternal treatments

Experimental groups were exposed to a daily, short stress session (NS, RS, MS) between 08h00–10h00 from day 11 to day 19 of pregnancy. Glucocorticoid and mineralocorticoid receptors are expressed in the developing rat brain on days 13–16 (Diaz et al., 1998). All maternal treatments, from the adaptation period until the end of pregnancy, are described in the protocol timeline (Figure 1).

Noise stress

Auditory stimulation was provided for 3 min per day, between 08h00 and 10h00 over a period of 9 days. Noise was generated by a HELLA twintone Horn, Japan, (110 dB output at 500 Hz, 12 V), fixed to the middle of the roof of the box. The sound pressure of 110 ± 3 dB (A) inside the box was confirmed with a sound level meter (Trotec B S06, France, measuring range: 40 to 130 dB and 31.5 Hz to 8 kHz). In order to

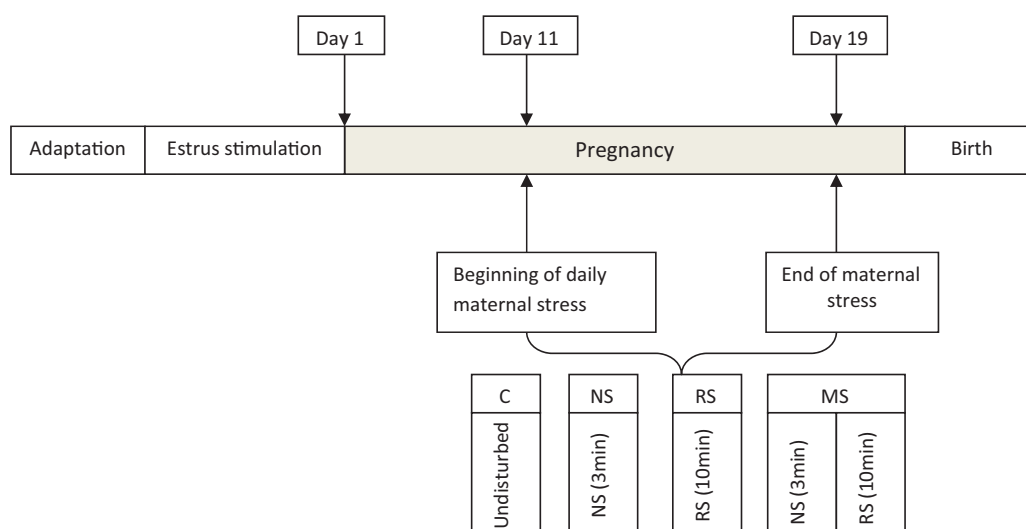


Figure 1. Timeline of the maternal treatment protocol. C (control group); rats of this group were left undisturbed. NS (noise stress group); females were exposed to daily sessions of noise stress for 3 min. RS (restraint stress group); females were restrained daily for 10 min. MS (mixed stress group); females were exposed to noise for 3 min, followed by 10 min of restraint. Stress sessions were administered daily from day 11 of pregnancy until day 19 between 08h00 and 10h00.

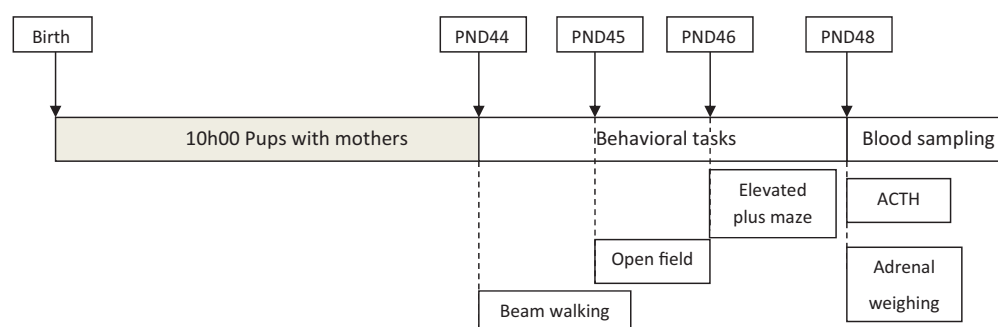


Figure 2. Timeline of the progeny treatment protocol. All male rats were kept undisturbed until postnatal day (PND) 44 when they began performing the following behavioral tasks between 08h00 and 10h00: the Beam Walking (BW) test on PND44, Open Field (OF) test on PND45, and Elevated Plus Maze test (EPM) on PND46. Blood samples were taken and adrenal glands were weighed on PND48 at 08h00.

assess the effects of prenatal exposure of high decibel noise on rats, it is a prerequisite that the frequency range of the stimulus should be within the range that the test species naturally communicates and hears. Here, the horn emitted a pure tone of 500 Hz. Noise was applied individually to rats placed in a box (20 cm × 30 cm × 30 cm) made of thick glass (1 cm) fitted with a ventilated top. **Noise induction was applied in a room separate from the colony room where rats were housed.**

Restraint stress

To induce chronic restraint stress, rats were restrained in individual opaque plastic cylinders (14.5 cm in length, 5.5 cm in diameter). The cylinder was fenestrated to allow equilibration of temperature and gases with the ambient air. Rats were restrained daily for 10 min, over a period of 9 days, between 08h00 and 10h00. During stress sessions, rats had no access to water or food.

Mixed stress combining noise and restraint stress

Rats were subjected to noise stress (as described) for 3 min, followed by restraint for 10 min (as described above) after a 5 min interval.

Neurobehavioral tests

Behavioral tests were performed in the following order: beam walking (BW) on PND44, open field test (OF) on PND 45 and elevated plus maze (EPM) on PND 46. All tests were conducted in a quiet room, following 20 min of habituation to the test room, during the light period (between 10h00 and 12h00). After behavior assessment, rats were returned to the colony room. All protocols of progeny manipulation are described in Figure 2.

Balance test: beam-walking (BW)

The runway used in the locomotor task was a wooden beam, 150 cm long and 4 cm wide on the running surface, elevated 60 cm above a table by a wooden support structure (Pisa, 1988a). Video recordings of beam running were recorded and analyzed in slow-motion playback. Two parameters were recorded: (1) time to cross the beam; and (2) number of mis-steps in each trial (either footfalls outside the beam or when a rat's paw extended over the narrow beam). The analysis was conducted blind to the group identity of the rats. Measures were also taken of the mean beam running time (in seconds) within each group. The beam walking test was

conducted on PND 44, before OF and EPM tests, because of its sensitivity to stress.

Open field (OF)

Following the model of Hall (1934), we used a device comprising a square-shaped floor (100 cm × 100 cm), divided by black lines into 16 squares of equal size (25 cm × 25 cm); 12 peripheral squares and 4 in the center. The floor was protected by 30 cm high transparent Plexiglas walls.

This test evaluates both locomotor activity (number of squares crossed, i.e. whenever a rat entered a new square with all four legs) and exploratory behavior (time spent in a corner versus time spent in the center for horizontal exploration, and time spent rearing on hind legs for vertical exploration). 20 min prior to test exposure, rats were moved to the test room for habituation, but remained in their home cages. After habituation, individual rats were placed into the center of the open field and their behavior was recorded for 5 min by video camera (Sony Steady Shot, Carl Zeiss, 14.1 MP, 26 mm). The apparatus was cleaned with 10% ethanol after each test.

Elevated plus maze (EPM)

The EPM test, which is a well-known paradigm for testing anxiety levels in laboratory rodents, consisted of two open arms (50 cm × 10 cm) crossed at right angles with two closed arms (50 cm × 10 cm, surrounded by 40-cm high Plexiglas walls). The apparatus was raised 50 cm above the floor. Rats were placed at the center of the maze, facing one of the closed arms and their behavior was video-recorded for 5 min. Rearing time and the time spent in both the open and closed arms were recorded. The percentage time in open arms [OAT % = (time spent in open arms × 100)/(closed arms time + open arms time)] and percentage time in closed arms [CAT % = (time spent in open arms × 100)/(closed arms time + open arms time)] were calculated. Rats spend less time in the open arms of the maze when they are afraid or anxious (Pellow & File, 1986). To minimize olfactory stimuli, the maze was cleaned with a 10% ethanol solution between sessions.

Blood sampling

On PND48, blood sampling was conducted in the early morning at 08h00. Terminal blood collection was performed from the venous sinus by penetrating the retro-orbital plexus with a capillary tube (Diehl et al., 2001), after ensuring that each rat was under deep general anesthesia with chloroform as inhaled anesthetic. A clean cotton ball was soaked in chloroform and kept under a desiccator with a tightly-closed lid into which each rat was placed. Depth of anesthesia was assessed by testing withdrawal reflexes after approximately 40 s following exposure. Blood collected in EDTA containers was used for analysis of ACTH plasma concentrations.

Plasma ACTH assay

Blood was centrifuged at 1730 *g* for 15 min and circulating concentrations of ACTH were evaluated using the Sandwich

Enzyme-linked Immunosorbent Assay according to the manufacturer's protocol (ACTH ELISA Kit, Sigma-Aldrich, USA). ACTH plasma concentrations were measured using a COBAS e 411 analyzer; assay detection limit was <1 pg/ml, intra-assay coefficient of variation was 6.7%.

Relative weight of adrenal glands

After blood collection, 16 rats in each group were euthanized and their adrenal glands were removed, trimmed of fat and weighed. Relative weight of adrenal glands was calculated as follows: actual adrenal weight/actual body weight (Pankakoski & Tähkä, 1982). Relative weight of adrenal glands was determined to investigate the chronic effects of our stress model on adrenal hypertrophy.

Statistical analysis

Data were analyzed using the statistical package Minitab® (version 16.0) to compare between-groups differences in ACTH concentrations and behaviors. Preliminary one-way ANOVA was performed to assess differential effects of prenatal stress. When a main effect of prenatal stress on ACTH, relative adrenal weights and behavioral variables was detected ($p < .05$), appropriate *post hoc* analysis was performed to evaluate significant differences. Dunnett's *post hoc* test was performed for comparison with the control group, and Tukey's test for multiple comparisons. All results are presented as the arithmetic mean ($\bar{X}$) ± standard error of the mean ($\bar{X} \pm \text{SEM}$). For all statistical analyses, a p value of less than .05 was considered as the significance level.

Results

Plasma ACTH concentrations

ACTH concentrations (Figure 3) were assayed on PND 48 in all groups. Preliminary one-way ANOVA revealed a significant between-group difference in plasma ACTH concentration ($p = .001$, $F(3,63) = 8.91$). Figure 3 shows that ACTH concentrations were highest in NS rats and significantly different from control (Dunnett's test: $p = .002$, $F(1,31) = 17.32$), and also high in RS rats and also significantly different from control ($p = .004$, $F(1,31) = 12.80$). In the MS group, plasma ACTH concentrations were significantly lower than for the NS group (Tukey's test: $p = .016$, $F(1,31) = 8.32$), but significantly higher than the control (Dunnett's test: $p = .034$, $F(1,31) = 2.79$).

Relative weights of adrenal glands

Statistical analysis revealed significant effect of prenatal stress on relative weights the adrenal glands ($p = .048$, $F(3,63) = 3.29$). Pairwise comparison using one-way ANOVA indicated that relative weights of adrenal glands only differed between NS and control rats (Dunnett's test: $p = .049$, $F(1,31) = 5.36$). Relative adrenal gland weights were not significantly different between either RS or MS versus control (Dunnett's test: RS vs Control, $p = .16$, $F(1,31) = 2.35$; MS vs

Control, $p = .132$, $F(1,31) = 2.82$). Hypertrophy of adrenal glands was only detected in the NS group. Actual body weights and actual and relative adrenal gland weights are given in Table 1.

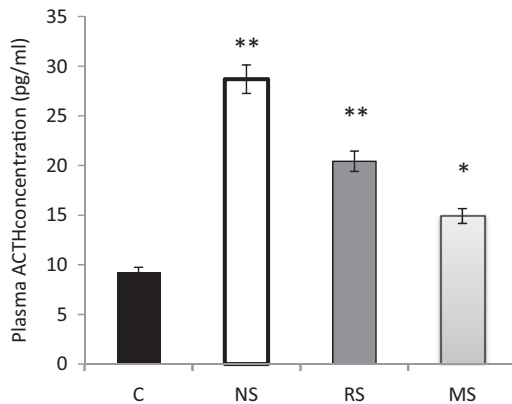


Figure 3. Effect of prenatal stress on ACTH concentrations (pg/ml) in control (C), noise stress (NS), restraint stress (RS) and mixed stress (MS) groups (expressed as Mean $\pm$ SEM) and compared by one-way ANOVA followed by Dunnett's *post hoc* test (Number of rats per group = 16). *Indicates p value is $< .05$ vs control. **Indicates p value is $< .01$ compared to control group.

Beam walking test

This test was performed on PND44. Between-group differences were observed in time spent to cross the beam ($p = .005$, $F(3,63) = 4.78$) and in the number of missteps ($p = .0001$, $F(3,63) = 15.74$). Results of the beam-walking task are summarized in Figure 4 and indicate that the NS group took a significantly greater amount of time to cross the beam than control (Dunnett's test: $p = .001$, $F(1,31) = 11.59$). ANOVA also indicated that the number of missteps was significantly higher for the NS group compared to control (Dunnett's test: $p = .0001$, $F(1,31) = 19.02$) (Figure 4(B)). No significant differences were observed in terms of time to cross the beam (Dunnett's test: $p = .79$, $F(1,31) = 0.07$) or for the number of missteps (Dunnett's test: $p = .56$, $F(1,31) = 0.32$) for the RS group compared to control. However, MS rats took longer to cross the beam than controls (Dunnett's test: $p = .021$, $F(1,31) = 6.22$), but did not differ from control in relation to the number of missteps (Dunnett's test: $p = .18$, $F(1,31) = 1.9$).

Interestingly, number of missteps in the MS group was significantly lower than in the NS group (Tukey's test: $p = .001$, $F(1,29) = 14.79$).

Open field test

There were significant difference among groups in all considered parameters (Time spent in corner: $p = .028$, $F(3,63) = 3.30$; Time spent in center: $p = .0001$, $F(3,63) = 7.93$; Number of crossed squares: $p = .014$, $F(3,63) = 3.85$; Rearing time: $p = .001$, $F(3,63) = 9.78$). Figure 5 illustrates the exploratory and locomotor performances of rats during the open field test. Panels A, C and D of Figure 5 reveal greatly decreased exploratory behavior in NS rats. Indeed, compared to the control group, NS rats revealed highly significant differences with (a) increased time spent in corners (Dunnett's test: $p = .0001$, $F(1,31) = 16.45$); (b) decreased time spent in the center (Dunnett's test: $p = .0001$, $F(1,31) = 19.55$ and (c) decreased rearing time (Dunnett's test: $p = .001$, $F(1,31) = 21.68$). For the RS group, only vertical exploration was significantly diminished compared to control (Dunnett's test: $p = .004$, $F(1,31) = 9.89$), with no significant differences for time spent in corners and/or in the center (Dunnett's test: $p > .05$). Compared to the control group, MS rats exhibited a significant decrease for both time spent in the center (Dunnett's test: $p = .018$, $F(1,31) = 6.20$) and rearing time (Dunnett's test: $p = .012$, $F(1,31) = 7.10$), and an increase in time spent in corners (Dunnett's test: $p = .037$, $F(1,31) = 4.83$).

Panel B of Figure 5 shows that locomotor activity was significantly lower in NS rats compared to the control group (Dunnett's test: $p = .006$, $F(1,31) = 8.65$). There were no significant differences in locomotor activity of RS and MS rats compared with the control group (Dunnett's test: RS vs Control: $p = .42$, $F(1,31) = 0.67$; MS vs Control: $p = .34$, $F(1,31) = 0.91$).

Elevated plus maze (EPM)

Preliminary ANOVA revealed significant changes in time spent in open arms ($p = .0001$, $F(3,63) = 14.15$), and time of rearing among groups ($p = .002$, $F(3,63) = 5.76$) (Table 2). In EPM tasks, anxiety levels are inversely related to the time spent in open arms (Estanislau & Morato, 2006). Figure 6(A) shows

Table 1. Body weight, adrenal gland weight and adrenal gland weight relative to body weight of the control and prenatally stressed groups.

	C	NS	RS	MS
Actual body weight (g)	206.8 $\pm$ 6.85	231.2 $\pm$ 9.47	226.8 $\pm$ 3.06	216.6 $\pm$ 8.63
Actual adrenal weight (mg)	45 $\pm$ 4	76 $\pm$ 4	69 $\pm$ 7	68 $\pm$ 8
Relative adrenal weight (mg/100g)	22 $\pm$ 4	33 $\pm$ 1*	30 $\pm$ 2.7	31 $\pm$ 2.8

Weights are presented as Mean $\pm$ SEM (Number of rats per group = 16) and were compared by one-way ANOVA followed by Dunnett's *post hoc* test. C: control; NS: noise stress; RS: restraint stress; MS: mixed stress. *Indicates p value is $< .05$ compared to control group.

Table 2. Proportions of time spent in open and closed arms during the elevated plus maze (EPM) task, expressed as mean (%) $\pm$ SEM and compared by one-way ANOVA followed by Dunnett's *post hoc* test ($n = 16$ rats per group).

	C	NS	RS	MS
% of time spent in open arms	39.2 $\pm$ 4.65	28.6 $\pm$ 5.28*	34.6 $\pm$ 5.02	18.3 $\pm$ 4.14***
% of time spent in closed arms	60.5 $\pm$ 4.78	74.6 $\pm$ 5.28	65.3 $\pm$ 5.02	81.2 $\pm$ 5.82

C: control; NS: noise stress; RS: restraint stress; MS: mixed stress. *Indicates p value is $< .05$, and ***indicates p value is $< .001$ compared to control group.

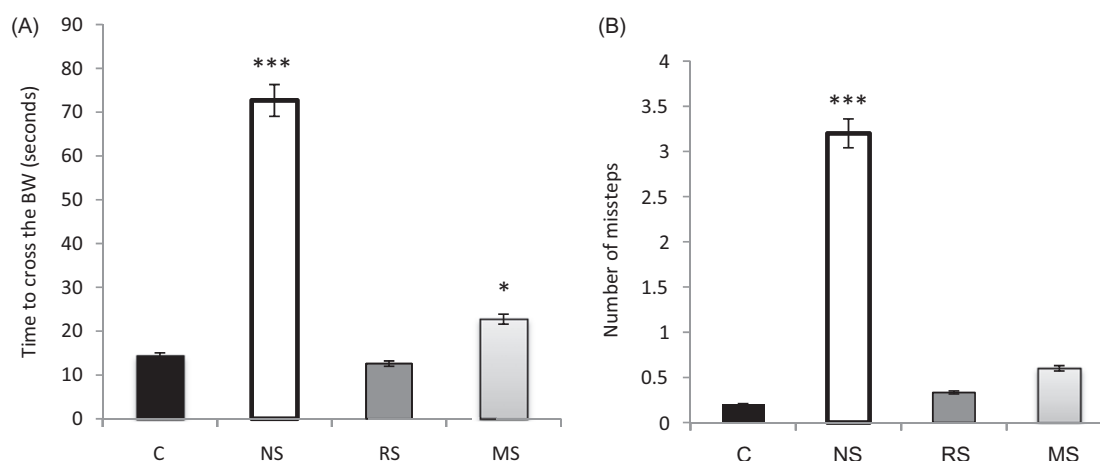


Figure 4. Effect of prenatal stress on time to cross (A) the beam and (B) number of missteps in beam walking test. Values are expressed as Mean $\pm$ SEM and compared by one-way ANOVA followed by Dunnett's *post hoc* test (Number of rats per group = 16). C: control; NS: noise stress; RS: restraint stress; MS: mixed stress. *Indicates p value is $< .05$ vs control. **Indicates p value is $< .01$ compared to control group.

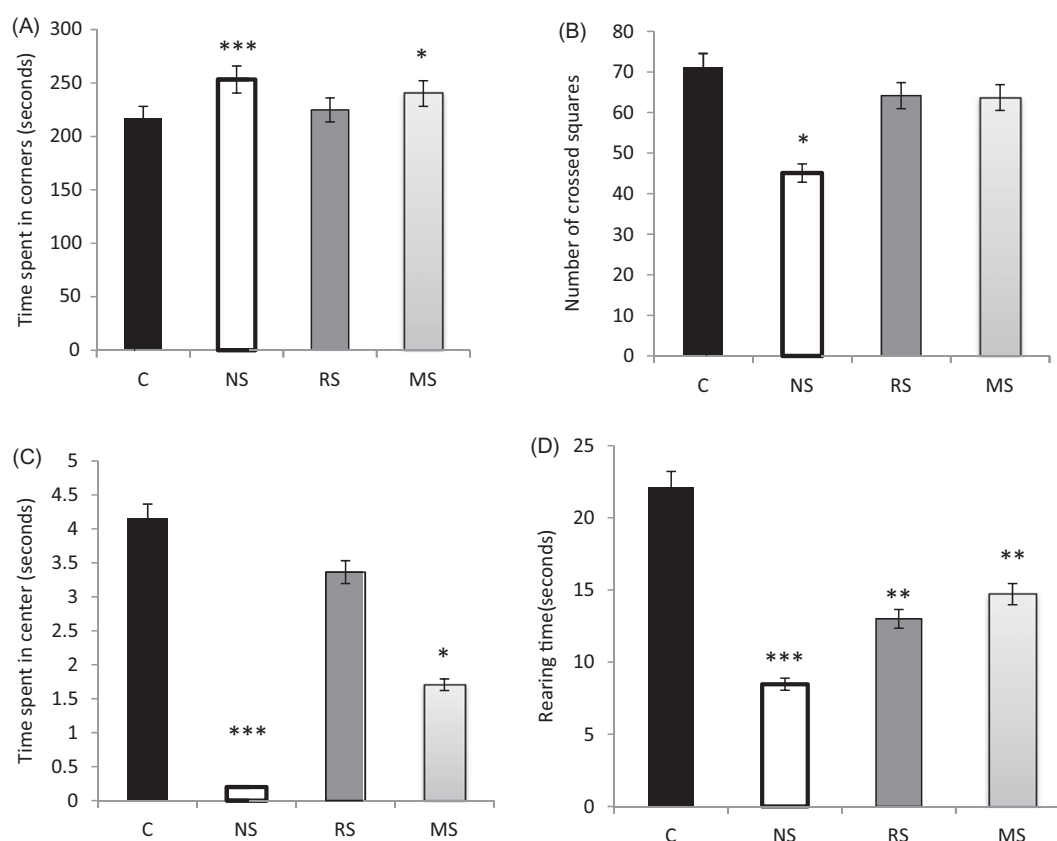


Figure 5. (A) Effect of prenatal stress on time spent in the corners, (C) time spent in the center, (B) number of crossed squares and (D) rearing time for the Open Field trials. Values are expressed as Mean $\pm$ SEM and compared by one-way ANOVA followed by Dunnett's *post hoc* test (Number of rats per group = 16). C: control, NS: noise stress, RS: restraint stress, MS: mixed stress. *Indicates p value is $< .05$. **Indicates p value is $< .01$ vs control. ***Indicates p value is $< .001$ vs control group.

that MS rats spent significantly less time in open arms than the control group ($p = .0001$, $F(1,31) = 16.15$), indicating that MS rats expressed high levels of anxiety. Although less pronounced, the NS group also exhibited higher anxiety compared to control (Dunnett's test: $p = .014$, $F(1,31) = 6.77$). However, time spent in open arms was not significantly different between the RS and control groups (Dunnett's test: $p = .12$, $F(1,31) = 1.97$). As shown in Figure 6(B), rearing time was significantly reduced in all stress groups compared to control, suggesting that prenatal noise (Dunnett's test:

$p = .0016$, $F(1,31) = 11.92$), restraint (Dunnett's test: $p = .022$, $F(1,31) = 5.74$) and mixed (Dunnett's test: $p = .045$, $F(1,31) = 4.36$) stresses all affected vertical exploration in adolescent rats.

Discussion

The goal of the present study was to determine the extent to which prenatal noise and restraint stress, applied individually, alter exploratory behavior, locomotor performance and

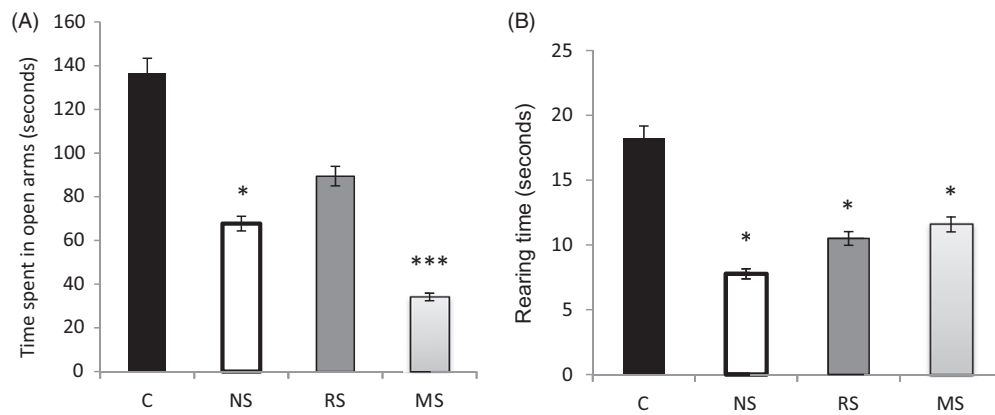


Figure 6. Effect of prenatal stress on (A) time in seconds spent in open arms and (B) time in seconds spent rearing in Elevated Plus Maze trials. Values are expressed as Mean $\pm$ SEM and compared by one-way ANOVA followed by Dunnett's *post hoc* test (Number of rats per group = 16). C: control; NS: noise stress; RS: restraint stress; MS: mixed stress. *Indicates p value is $<.05$. **Indicates p value is $<.01$ vs control. ***Indicates p value is $<.001$ vs control group.

balance control functioning in juvenile male rats, and how these stressors interact to modify behavior when applied together.

Our results report alterations of locomotor and exploratory behaviors, as well as endocrine disruptions, induced by individually and prenatally applied stress. We found that a combination of noise and restraint stress reduced these effects in rats, but did amplify anxiety levels. Balance control was highly affected by repeated prenatal exposure to noise, which was not the case for restraint alone or combined stress. Previous data have shown that exposure to loud noise might affect not only hearing regions of the brain (Kraus & Canlon, 2012), but also the amygdala and hippocampus that receive direct and indirect neuronal inputs from the central auditory system (Munoz-Lopez et al., 2010; Winer & Lee, 2007).

It is generally assumed that the vestibular system, which plays a key role in posture control, sends signals through the vestibular nerve fibers to the vestibular nuclei (Cullen, 2014) to maintain body posture. Disruption of normal vestibular functioning can result in direct oscillopsia (during walking or standing) or alterations of balanced gait (Curthoys & Halmagyi, 1995; Etienne & Jeffery, 2004). Additionally, it appears that vestibular dysfunction involves a more complex syndrome characterized not only by balance deficits, but also by anxiety disorders (Smith et al., 2010). We found that anxiety level (assessed according to EPM performance) was markedly increased in prenatally noise-stressed and mixed stress rats compared to restraint-stressed and control rats. The vestibular information provided in the ascending pathways to the limbic system and neocortex is required for normal spatial orientation (Chen et al., 2013; Smith & Zheng, 2013) and, in the absence of this information, internal representations become inaccurate and ambiguous, and cognitive performance is affected.

Furthermore, the striatum has been shown to be involved in sensori-motor activity. Pisa and colleagues (Pisa, 1988a, 1988b) reported that localized lesions in the lateral striatum decreased walking speed and balance gait during beam-walking. Based on these and our results, we suggest that disturbed balance may be caused by local alterations to specific brain regions, such as the vestibular system or

striatum or other, by a specific stressor during intrauterine life. Further investigations are needed to determine which brain regions are involved in impaired balance control arising from prenatal noise stress. Previous studies (Jensen Peña et al., 2012) suggest that this impairment might be mediated by maternal glucocorticoids (GC), which can have free access to the entire fetal nervous system if stress exposure downregulates restrictions to the transfer of maternal GC to the fetus.

We also noted that prenatal restraint stress had no effect on balance compared to control. This result supports the idea that the nature of a stressor selectively and differently affects brain regions, with consequent differential effects on behavior and other functions potentially linked to higher reactivity of the hypothalamic–pituitary–adrenal axis (Weinstock, 2008). Interestingly, the combined stress rats took a statistically equivalent amount of time to cross the beam in the BW test as control rats. Thus, perhaps exposure to prenatal noise followed by prenatal restraint reverses the balance impairment observed in rats that only experience prenatal noise stress. To our knowledge, this is the first time that the impact of such a combination of stressors on balance control has been described.

Reduced crossing events in OF trials suggest that locomotor activity was most impaired in noise-stressed rats. Similarly, exploratory behavior (measured as rearing time for vertical exploration and time spent in the center/corners for horizontal exploration) was greatly reduced in the noise-stressed group compared to the restraint-stressed and mixed stress groups. The reduced locomotor and exploratory behavior of stressed rats confirms previous reports showing that stressful environmental factors occurring during perinatal life affect brain plasticity, as well as physiological and neurological functions (Maccari & Morley-Fletcher, 2007), such as down-regulation of the dopaminergic activity that is involved in motivational behaviors (Keck et al., 2013; Powell et al., 2003). Curiously, combined prenatal application of noise and restraint stress did not result in the significant alterations of locomotor and exploratory parameters observed when noise and restraint were applied separately. This result suggests that neurobehavioral responses to multiple stressors do not

necessarily imply additive effects. Evans (1984) suggested that some environmental stressors applied in combination can have a lesser impact on subjects than expected from their individual application. Attenuation of the negative effects (on locomotion and exploration) did not apply when anxiety levels were examined, which were amplified in combined stress rats (revealed by EPM trials). Indeed, our EPM trials revealed that anxiety levels were increased in all prenatally stressed rats compared to controls, notably excepting those subjected to restraint stress. This result is consistent with many previous studies reporting that prenatal stress enhanced the anxiety or depression-like behavior in rats (Maccari & Morley-Fletcher, 2007; Weinstock, 2008).

Though there are many pathways through which maternal stress during pregnancy can shape offspring neurodevelopment, stress-induced elevations in maternal GC levels have emerged as a primary mechanism (Jensen Peña et al., 2012), supported by studies in rats in which the effects of prenatal stress are attenuated in adrenalectomized females (Barbazanges et al., 1996). In our study, the low anxiety in the restraint stress group might be explained if 10 min of maternal restraint per day was insufficient to induce an effect.

However, we found mixed stress rats to be highly anxious compared to both the noise and restraint stress groups. It seems that a combination of prenatal noise and restraint stress induced additive effects, but only for anxiety levels. Taken together, we can deduce that prenatal exposure to combined noise and restraint stress simultaneously had a reduced impact (non-additive effect) on locomotor and exploratory activities compared to individually applied stresses, but had an enhanced impact (additive effect) on anxiety levels. Thus, our findings suggest that stressors (monotypic or multiple) induce differential effects when they are applied together or separately, abrogating some behavioral disruptions or enhancing others. We propose further studies to clarify how multiple stressors can combine to attenuate or potentiate their effects.

ACTH concentrations and relative adrenal weight data support our findings. ACTH concentrations were not significantly higher in rats that underwent mixed prenatal stress, but noise-stressed rats exhibited significantly increased ACTH concentrations. Similarly, hypertrophy of adrenal glands was only registered for the noise-stressed group, suggesting that repeated loud noise, even applied just for 3 min/day, elicits the strongest effect as hyperactivity in the HPA axis. The mechanisms involved in interactions between stressors are still unknown. To elucidate the relationships between prenatal stress and the different effects observed here, further study of maternal behavior under the same experimental conditions will be necessary. Such studies should examine maternal GC levels and the impact of giving anxiolytic drugs to pregnant rats to determine if diminished anxiety can influence the subsequent behavioral alterations in offspring.

Conclusions

In summary, results presented here provide additional evidences on deleterious effects of prenatal exposure to stress on locomotion and exploratory behavior as well as the

enhancement of anxiety like behavior. The main findings of this work are (a) The impact of the nature of stressor in yielding different biological and neurobehavioral effects. Here, prenatal noise stress was the stressor that generated the stronger alterations of locomotion, exploration and reactivity of the HPA axis (high plasma concentration of ACTH and adrenal hypertrophy); (b) the exposure to combined stressors can generate at the same time, additive and non-additive effects comparable to those observed with separate application. Locomotion, horizontal exploration and balance impairments were attenuated, and anxiety level was enhanced by mixed stress; (c) finally, locomotor activity was highly affected only by noise and not restraint stress suggesting that the two stressors interact differently with brains regions related to balance control. However, the mechanisms involved in this alteration remain unknown and data obtained by a behavior assessment battery have a limited relevance for understanding the origin of neurobiological abnormalities observed, and further research is needed to clarify how multiple stressors interact to induce at the same time an increase of some alterations and decrease of some others.

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Disclosure statement

The authors of this paper declare no conflict of interest.

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